

Optimizing Delivery Cost and Time Using Real-World Data



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This project focuses on solving a **multi-objective optimization problem** in logistics. The objectives are to minimize **cost, delivery time, and emission** using real-world data. Methods such as Scalarization, Greedy, NSGA-II and PSO are compared to find the best trade-off solution.

Introduction & Problem Statement

- Logistics systems require optimization of multiple conflicting objectives
- Objectives:
 - Minimize Cost
 - Minimize Time
 - Minimize Emission
- Trade-off:
 - Faster delivery → Higher cost & emission
 - Lower cost → Higher time

Objective & Dataset

Objective:

- Model real-world delivery optimization
- Minimize cost, time, emission
- Compare optimization methods
- Analyze trade-offs (Pareto front)

Dataset:

- Brazilian E-commerce (200 samples)
- Features:
 - Cost, Time
 - Distance, Weight, Zone
- Emission:

$$e_i = 0.02d_iw_i + \frac{5}{t_i + 1}$$

Decision Variable:

$$x_i \in \{0, 1\}$$

Methods Used:

- Scalarization (Weighted sum)
- Greedy (Cost-based)
- NSGA-II (Evolutionary)
- PSO (Swarm-based)

Objective:

$$\min \sum (w_1 c_i + w_2 t_i + w_3 e_i) x_i$$

Constraints:

$$\sum x_i \leq 50$$

$$\sum w_i x_i \leq 300$$

NSGA-II Convergence

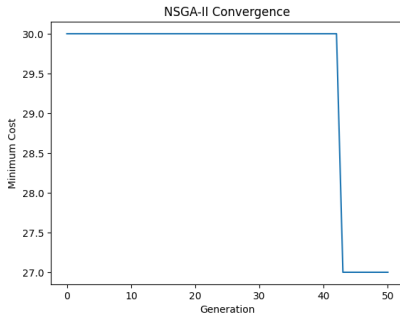


Figure: Minimum Cost vs Generations

- Cost decreases over generations
- Algorithm converges to optimal solution
- Stable curve indicates convergence

Conclusion: NSGA-II shows efficient convergence behavior.

Pareto Front (Cost vs Time)

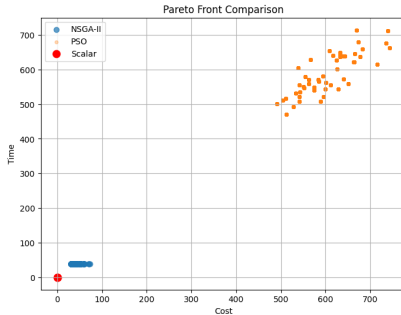


Figure: Comparison of NSGA-II, PSO and Scalarization

- NSGA-II provides low cost and low time solutions
- PSO shows higher cost and time
- Scalarization gives only one solution

Conclusion: NSGA-II provides best trade-off solutions.

3D Pareto Front

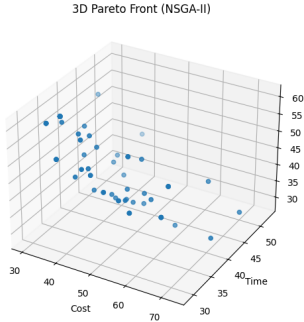


Figure: Cost, Time and Emission Trade-off

- Shows trade-off among cost, time, emission
- No single optimal solution exists
- Improving one objective worsens others

Conclusion: Multi-objective problems require trade-off solutions.

Results Comparison

Method	Cost	Time	Emission
Scalarization	0	0	0
Greedy	442.71	402	24669.77
NSGA-II	30	29	28
PSO	491	501	509

- NSGA-II provides best balanced solution
- Uses non-dominated sorting and crowding distance to efficiently handle multiple objectives and find optimal trade-offs

- Interactive Streamlit application
- Allows weight adjustment and visualization
- Same result issue due to dominant solution and scalarization with limited dataset

Open Web App

Limitations & Improvements

- Small dataset (200 samples)
- Simplified emission model
- Improvements:
 - Larger dataset
 - Real CO2 model
 - Hybrid algorithms (NSGA-II + PSO)

Conclusion

- Multi-objective optimization requires trade-offs between cost, time, and emission
- NSGA-II provides the best balanced solutions (Pareto optimal)
- Emission represents environmental impact and can be reduced by efficient routing, load management, and balanced delivery time

- Real-time logistics integration
- Deep learning / reinforcement learning
- Smart delivery systems

Thank You!

Questions?